

Lecture 01

Introduction - Course Welcome

Cyber Security and Privacy | NPTEL 106106248

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Video: <https://www.youtube.com/watch?v=OYsY5B9pqYU>

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Hello and welcome to this course on cyber security and privacy. This course is about the dark side of the cyber world. Cyber technologies or the cyber world has emerged extensively in the current times. In the contemporary world we all use technology, digital technologies.

It's not just human beings in as one entity, but people in groups, people in organizations, people in government, we use technology in different contexts. And the technology has huge potential to ease our life and also increase efficiency and effectiveness of work. But at the same time there is a dark side to it. As the adoption of technology grows, so are the challenges that are facing the use of technologies, especially from the dark world.

Hackers get updated on the new technologies. And they try to disrupt, attack, and cause damage to technology for various reasons, okay. So unless you secure your technology, unless you secure your digital assets, your entire business or your entire organization may collapse if you are very much dependent on digital technologies. Recent cases or recent incidents throw light on the potential of cyber attacks on disrupting and destroying the digital world.

We increasingly hear about drones and how they destroy assets, not only physical assets, but even human beings. We also hear about other kinds of cyber attacks, threats that are increasing in the cyber world. And the purpose of this course is manifold. One is to generate awareness about the dark side of the digital world, especially for practicing managers and cybersecurity professionals.

It is important to understand what are the cybersecurity challenges. The other aspect of the course is that this course deals with cybersecurity and privacy not from a technological point of view alone, but it also looks at cybersecurity as a management issue, as a governance issue. So, governance risk and compliance are three critical aspects of cybersecurity from a managerial perspective. So, this course takes the administrative or managerial perspective and also looks at technology that is challenged or that can be destroyed and technology is also seen as a source for protecting technologies.

So, the twofold role of technology would be dealt with in this course. Okay, there is another aspect to this course, which is pure management of cyber assets. So, and secure securing cyber assets. So, for that actually we go to fundamental lessons of management where planning plays an important role.

So, we look at cybersecurity planning from two perspectives. One is the contingency planning where the basic assumption is that things can go wrong. And incidents can happen. Cyber attack can happen.

Somebody can take control of your machine and ask for money, you know, the ransomware

attacks which are growing increasingly today. So, in this kind of context when an attack happens, how can an organization get its technology restored to a normal condition? How fast it can recover? So, that is the contingency planning and impact analysis, etc.

Okay, that is one aspect of planning. The other aspect of planning is to look at cyber assets as assets to be protected. And therefore, you actually take a risk management perspective, evaluate the value of each of the assets and what are the potential threats that can actually happen and what what are the probabilities and you try to arrive at certain quantitative or qualitative estimate of so-called residual risk. And then plan management action based on what should be done for each asset.

So, that is actually a planning based on risk management, okay, risk perspective. The other is So, here we don't assume that something has already happened, but the it is more it is not reactive, but more preventive in nature. Okay, so, we this course deals with both the aspects of cybersecurity planning. And it also covers certain standards that are useful in implementing this kind of plans.

And then subsequently, this course also deals with cybersecurity technologies, especially from a protection point of view. What are the technologies that are available to protect cyber assets? So, in that context, we will be discussing cryptography and recent developments in cybersecurity technologies. We will also be having industry expert to share current technologies that are at use for cybersecurity.

So, alongside, another aspect of this course is information privacy. So, we move from cybersecurity to information privacy. Because often times what is at risk is data. Okay?

It is the data that the hackers often steal. Okay? So, there is a value to individuals data. Okay?

So, we build the concept of privacy from fundamentals. What is privacy? What is information privacy? And what are the current developments that are related to information privacy worldwide?

We hear about regulations that are enacted in different parts of the world in the context of growing use of information technology by individuals and organization and government. So, we see in India, we are waiting for an act called digital personal data protection act, okay, called DPDP. Which is expected to become a law very soon. So, Indian government, you can see is very conscious of the importance of information privacy of individuals.

The The Supreme Court of India upheld privacy as a fundamental right. So, these are current developments, you know, as technology becomes pervasive, the vulnerability of individuals grow and it becomes the responsibility of governments to protect it. And how do you actually protect privacy? It is using technology.

So, you see the role of technology or cybersecurity is also to protect information privacy. And then, who is responsible for it? What are the Who are the different stakeholders involved? And what do what what does regulation do in different parts of the world right from Europe GDPR and India's DPDP and so on.

And so we we also look at cybersecurity and technology from managerial, economic, and

strategy perspective. Okay? So, who should take this course? Those who use technology and those who are responsible for technology and who are over use technology today would benefit from this course.

So, warm welcome to this course on cybersecurity and privacy. Thank you.